

# Serious Shea Senegal : Business Plan for the Great Green Wall

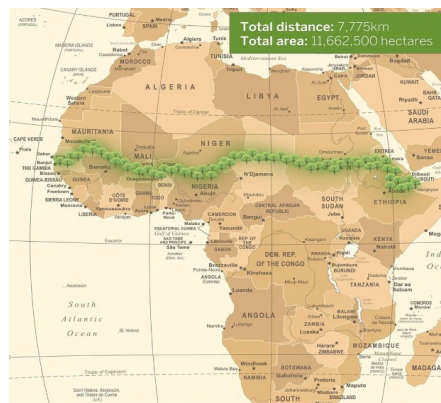
Prepared by William Kwende, Dr. Sarah Jones (Clarity Secretariat) & Sencer Karasu

With grateful thanks for their feedback to Simon Jones (Partner, Cognis Capital); George Sycip (Chairman, Give to Asia); Prof. Michael Mainelli (Chairman, Z/Yen Group); Uplink- World Economic Forum (Investors Group). The final plan does not necessarily represent the reflections of these advisors.

This Business Plan for Serious Shea Senegal was adapted and evolved from a Business Plan for Serious Shea Burkina Faso founded by Golden Organics, Impact Agri & the Women's Shea Butter Union & funded by the World Bank, the African Development Bank & the Burkina Faso Government.

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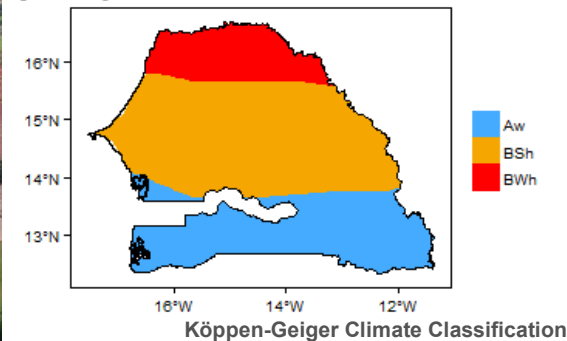


# 1. INTRODUCTION

Since 2007 Africa has been united in building the Great Green Wall — an 8,000 kilometer strip of reforested land across the continent from Senegal to Djibouti and the biggest living natural wonder in the world. The initiative is planting trees along this 3 million km<sup>2</sup>, semi-arid ecological transition zone, called the Sahel, to act as a buffer to prevent the Sahara Desert from expanding south. Combating desertification and land degradation will enable the local people to work their lands, providing food security, jobs, and a reason to stay for the millions who live along this tree wall. Within the Sahel Region, the Great Green Wall Initiative is initially focusing on the most urgent areas that are at highest risk of desertification or have the highest potential for preventing the spread of desertification while at the same time improving conditions for the local communities.

A decade since starting The Great Green Wall initiative is only 15% underway. Support from within Africa and the World is urgent for climate resilience, for nature and for communities. The transition to clean energy is critical so the trees planted do not end up as charcoal from burning wood fuel for food processing and in kitchens and so added value processing remains in the region to boost rural economy with action to protect the native forests.

## SENEGAL



Population: 18 million people  
Area: 19672 km<sup>2</sup>  
Climate: Aw Tropical savanna  
Bsh Warm semi-arid  
Bwh Warm dessert

GDP: US\$ 27.7 Billion  
Employment: 60% Food Crop Production  
(World Bank Statistics @ latest available date)  
Rural Energy Access 38% Un SE4all-africa.org

The current reliance of wood fuel in rural communities is such that e.g. 20kg wood are used for every 1kg of shea butter processed from shea fruit and nut collection.

Serious Shea is Smoothing the Way to a Better Future | Green Solutions | WEF <https://youtu.be/02ZuQH6-5JA>



The Great Green Wall Regeneration, Deforestation-free, Zero-carbon Value Chain Business Model as proposed by Serious Shea Senegal, won the United Nations and World Economic Forum endorsed TRILLION TREES SAHEL and GREAT GREEN WALL CHALLENGE in 2021 as a TOP INNOVATOR which is leading the way.

After 2 years of feasibility studies, technical preparations, and the building of linkages between the private sector, local communities and the public sector, Serious Shea is ready to launch its Great Green Wall Challenge starting from Senegal in the western most part of the Sahel.

Serious Shea Senegal's innovative approach is:

- Science based;
- Socially inclusive;
- Partnership based;
- Built on local skills development;
- Well managed with people and technology.

Management technologies include clean energy processing, Circular Economy waste recycling for biogas and soil conditioners, soil water retention gels, QR coded supply chain, planet.org/nicfi satelliteland-use & carbon information, drone data verification, Chain Zy (aka Blockchain) time stamp so financial and ESG information cannot be altered.

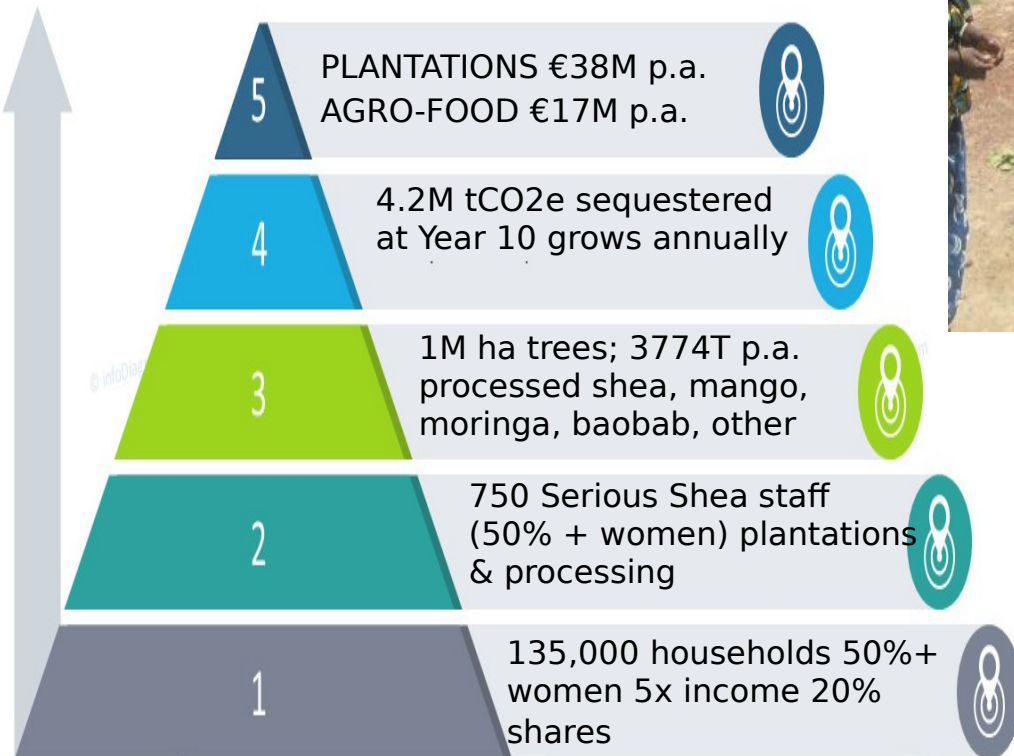


## 2. OVERALL ACTIVITY

### PLANTATIONS & DEFORESTATION-FREE ZERO-CARBON AGRO-FOOD VALUE CHAIN

Nurseries & Native Forest Plantations  
Clean Energy Processing for shea butter, dried mangos, baobab and moringa powder and oils

€ 25 M investment  
€ 50 M p.a. gross profit Year 10;  
40% Carbon Equity; 60% Debt



### 3. OBJECTIVES

#### Overall Objective

Implement, starting in Senegal, not only a profitable, but also a socially and environmentally sustainable business model, which won the United Nations and World Economic Forum endorsed Trillion Trees Sahel and Great Green Wall Challenge in 2021. The three pillars of this innovative business model are:

- (1) a regeneration strategy;
- (2) deforestation-free zero-carbon agroforestry processing value chain;
- (3) women and youth group partners.

#### Working Objectives

(1) Plant and protect 35M native trees by 2030 with an 80% survival rate which sequester 1 tCO<sub>2</sub>/year each at maturity and contribute to climate change adaptation by greening and holding back Sahara desert expansion from the north:

- 5 nurseries each producing 1 million native tree seedlings;
- 0.9 M ha of primary native forest plantations (10% of which are no harvest soil quality & biodiversity reference areas);
- 0.1M ha secondary plantations of native tree corridors & fences with crops such as moringa.

(2) Transition from firewood energy and create added economic value which stays in the region by building and operating 26 facilities in three phases within 6 native forest plantation zones:

- 18 multi-use clean energy processing, quality control and true value chain data centres for processing forest fruits such as baobab and mango and agrofood products such as moringa;
- 2 clean energy shea butter processing clusters (2 regional centres each serviced by 3 collection centres);
- a circular economy recycling 100% bio-waste from the 26 processing units to produce biogas fuel for processing and households, liquid fertilizer and soil conditioners (biochars and compost).

(3) Achieve financial and business governance inclusion of women and youth groups through co-operative instruments which also give land ownership rights by:

- Increasing the income of 13,500 cooperative workers at least 5\* the minimum wage (US\$166.65/year as 23.03.23)
- creating a Community Fund which achieves the minimum global estimate for all the households (135,000 people) to achieve the Sustainable Development Goals #3 Universal Health Care; #4 Primary and Secondary Education; #6 Clean drinking water and sanitation by 2030.

## 4. BUSINESS DESCRIPTION

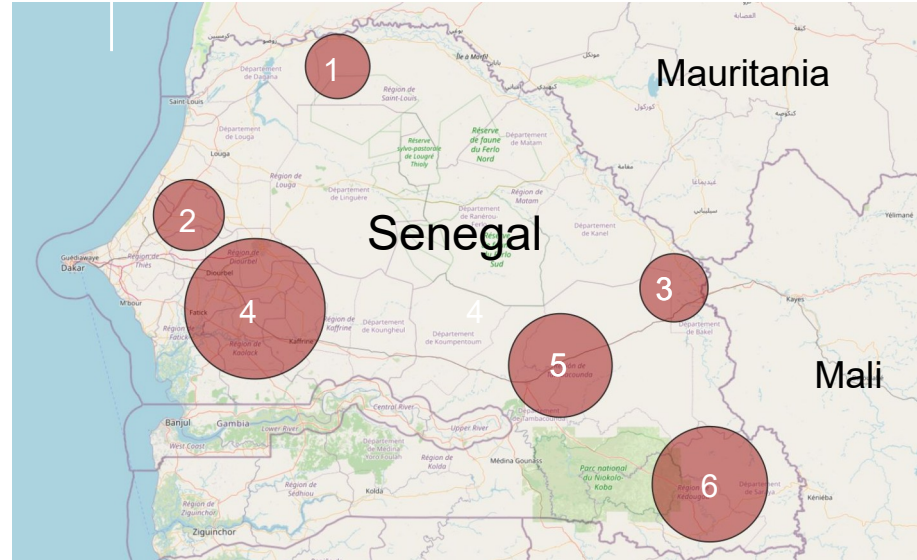
### 26 Facilities: 20 Processing within 6 Plantation Zones which include 5 Nurseries

#### PHASE 1 (2024-2025)

| Zones         | Z 1  | Z2 | Z3   | Z4 | Z5 | Z6 |
|---------------|------|----|------|----|----|----|
| Nurseries     | 1    | 1  | 1    |    |    |    |
| Industry      | 1    | 1  | 1    |    |    |    |
| Trees Planted | 0.5M | 1M | 0.5M |    |    |    |

#### PHASES 2&3 (2026 onwards)

| Zones         | Z 1  | Z2   | Z3   | Z4    | Z5   | Z6    |
|---------------|------|------|------|-------|------|-------|
| Nurseries     |      |      |      |       |      | 2     |
| Industry      | 1    | 1    | 1    | 2     | 1    | 6     |
| Trees Planted | 3.7M | 3.8M | 1.5M | 18.7M | 1.5M | 10.1M |



**Trees Planted\***      **35 M**

**Total Hectares**      **1 M**

\*@80% survival rate



## 4. BUSINESS DESCRIPTION

### Market Analysis of Value Chain

| Tree           | Latin name                  | Value chains considered            | Inter-national market potential | National/regional market potential | Environmental impact potential | Social impact potential |
|----------------|-----------------------------|------------------------------------|---------------------------------|------------------------------------|--------------------------------|-------------------------|
| Baobab         | <i>Adonsonia digitata</i>   | Fruit powder, oil, leaf powder     | Very high potential             | Some potential                     | High potential                 | Very high potential     |
| Balanites      | <i>Balanites aegyptiaca</i> | Oil                                | Good potential                  | High potential                     | High potential                 | Low potential           |
| Moringa        | <i>Moringa oleifera</i>     | Dried leaves/tea, leaf powder, oil | High potential                  | High potential                     | High potential                 | Very high potential     |
| Neem           | <i>Azadirachta indica</i>   | Oil, leaf powder                   | High potential                  | Some potential                     | Very high potential            | Low potential           |
| Kinkeliba      | <i>Combretum micranthum</i> | Dried leaves/tea                   |                                 | Very high potential                | Some potential                 | Some potential          |
| Locust beans   | <i>Parkia biglobosa</i>     | Seeds                              |                                 | High potential                     | High potential                 | Low potential           |
| Tallow tree    | <i>Detarium senegalense</i> | Fruit juice                        |                                 | Very high potential                | High potential                 | High potential          |
| Shea           | <i>Vitellaria paradoxa</i>  | Butter                             | Very high potential             | Very high potential                | High potential                 | Low potential           |
| Acacia senegal | <i>Senegalia senegal</i>    | Gum arabica                        | Very high potential             | No potential                       | Very high potential            | No potential            |
| Acacia seyal   | <i>Vachellia seyal</i>      | Gum arabica                        | Very high potential             | No potential                       | High potential                 | No potential            |

#### Key

 Very high potential  
 High potential  
 Good potential

 Some potential  
 Low potential  
 No potential

## 4. BUSINESS DESCRIPTION

### 5 Nurseries, 6 Plantation Zones

#### 1. Nurseries

5 nurseries each growing 1 M trees  
*Accasia mellifera*; baobab, tallow tree, shea, *Acacia senegalese*, ballenite & moringa saplings



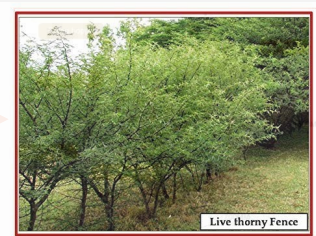
#### 2. Primary Plantations

5 Primary Plantations 0.9M ha  
*Accasia mellifera* (fences); baobab, tallow, shea, *Acacia senegalese*, ballenite



#### 3. Secondary Plantations (agroforestry)

0.1 M has forestry with moringa, *Accasia mellifera* fences.



## 4. BUSINESS DESCRIPTION

### 18 Multi-funcional Clean Energy Processing Centres; 2 Shea Butter Processing Clusters

#### 1. Collection & Boiling

*Solar cookers and steam boilers are used to process for oils and shea butter, eliminating the need for inefficient traditional wood-burning alternatives*



#### 2. Drying & Crushing

*Mechanical crushing and milling equipment significantly reduce the physical burden on the women's groups and increase their productivity*



#### 3. Cooking & Extraction (Shea)

*Gas stoves, shea butter extraction apparatuses and food-grade packaging equipment prepare the shea butter for export*





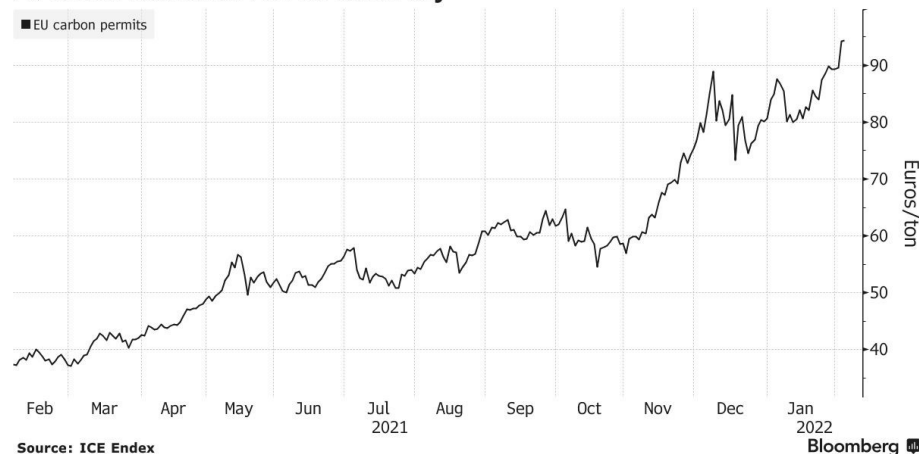
## 4. BUSINESS DESCRIPTION

### Combined Revenues from Clean Energy Processing and Planting Trees

| € M                   | 2023     | 2024       | 2025       | 2026        | 2027        | 2028        | 2029        | 2030        | 2031        | 2032        | 2033        |
|-----------------------|----------|------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Revenue from Industry | 0        | 0.5        | 5.0        | 7.9         | 11.6        | 16.0        | 16.3        | 16.6        | 16.8        | 17.1        | 17.1        |
| Revenue from Carbon   | -        | -          | 0.2        | 0.9         | 2.6         | 5.5         | 9.6         | 14.3        | 19.7        | 30.0        | 37.9        |
| <b>Total revenue</b>  | <b>0</b> | <b>0.5</b> | <b>5.2</b> | <b>8.80</b> | <b>14.2</b> | <b>21.5</b> | <b>25.5</b> | <b>30.9</b> | <b>36.5</b> | <b>47.1</b> | <b>55.0</b> |

#### Carbon Keeps Going

EU carbon allowances rise for fourth day



## 4. BUSINESS DESCRIPTION

### Financing and Marketing through Partners

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#### Financing Partners



#### Agro-food value chain partners



#### Customers

Business and Industry (carbon offsets)

Imisk food distribution channels

Natural cosmetics (L'occitane, Body Shop and other cosmetics groups through existing market relations and surveys by Agritech Group & Serious Shea Burkina Faso)

#### Target Audiences

Certified Carbon Offsets Worldwide

Heath Foods (bulk for retail) Middle East, US, Europe

Shea butter and oils (bulk for retail) Middle East, US, Europe



## 5. GOVERNANCE

### Shareholders



**60% SHARES; GOLDEN ORGANICS GLOBAL (20% FOR A COMMUNITY FOUNDATION & FOR RE-INVESTMENT TO SCALE THE BUSINESS MODEL THROUGH-OUT THE GREAT GREEN WALL)**

William Kwende, CEO Golden Organics & Serious Shea Executive Chairman is also Founder & CEO of Agritech Group which has organized the collection of shea nuts and other African commodities for the major international markets in Europe and Asia since 2010 (e.g 5-10 000 T /year shea nuts). Agritech Group has processed and exported shea butter for several years, produced and planted millions of trees in arid lands, and successfully certified a carbon generating project. William has a masters in Business Planning from Quantic School of Business and Technology, and was joint author of the Serious Shea Project which won the WEF-Uplink 1t.org Sahel and Great Green Wall Challenge in 2021. William has since played a major international role to advance the GGW initiative including at DAVOS 2022, with the African Union, the European Parliament, the UNFCCC and with public and private investors.



**SENEGAL WOMEN'S & YOUTH GROUP COOPERATIVES 20% GOLDEN ORGANICS SHARES HELD IN COMMUNITY FUND TO BE SET UP AS A FOUNDATION**

Fatou Mboup, Team Leader for Senegal Cooperatives and Value Chains, specialised in communications and marketing at Nice Cabuccia Visa International College (France). She has 20 years experience in communications and public relations including in companies such as Canals Plus Horizons, MATFOURCE and ABM.



**40% SHARES: CARBON EQUITY MANAGED BY MIROVA SUSTAINABLE INVESTMENT FUND**



## 6. IMPACT

### By SDG Indicator

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Private financial flows to plant and protect native forest are increased by € 30M by 2030 planting and protecting 0.9M ha native forest, 35 M diverse native trees, reducing 1 M ha degraded land and creating 10% soil quality and biodiversity reference areas in preparation for potential soil carbon and biodiversity credits, for pollinators, and to help maintain genetic diversity.  
(SDG 15.1.1; 15.3.1, 2.4.1)



€ 30M is mobilised to mitigate and adapt to climate change, sequestering 4.2 tCO<sub>2</sub>e by 2033 and substituting CO<sub>2</sub> from industry by replacing firewood fuel with clean energy (to quantify).  
(SDG 13.2.2; 13.a.1; 9.41)



100% clean energy (solar and biogas) powers 26 agro-food processing facilities linked to 14,250 jobs (Serious Shea employees and Co-operative workers), powers households for 135,000 people and creates gross revenues of € 30M by 2030.  
(SDG 7.11; 7.12; 7.31; 7a1)

## 6. IMPACT

### By SDG Indicator



3774 M T p.a. food is processed to reduce food loss to zero in processing and minimised from a regional loss of 80% including storage and transport to wholesale.  
(SDG 12.3.1)

1.0M ha and 3774 M T p.a agro-food are produced with sustainable practices which include water saving technologies, deforestation-free, zero-carbon, zero-waste processing.  
(SDG 2.4.1)



Income to 135,000 households is increased by at least 5\* the minimum wage (US\$ 117/year) and 1.5\* the average wage of the largest small holder farmers (3Ha US\$565/year ).  
(SDG 1.1.1; 2.3.2)

13,500 Co-operative workers, at least 50% women have equal payment and land rights secured through co-operative instruments.  
(SDG 5a1, 10.1.1)



Individual income increase is at least doubled in a community fund to provide minimum financial flows estimated to achieve Universal Health Care; primary and secondary education for all, clean drinking water and sanitation (est. US\$1100- US\$1340 /household).  
(SDG 3.8.1; 4.1.1; 6a1; 6.2.1)





## 7. FINANCIAL HIGHLIGHTS

### P&L & Operating Assumptions

| € M                     | 2023         | 2024         | 2025         | 2026       | 2027        | 2028        | 2029        | 2030        | 2031        | 2032        | 2033         |
|-------------------------|--------------|--------------|--------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|
| Revenues from Industry  | 0            | 0.5          | 5.0          | 7.9        | 11.6        | 16.0        | 16.3        | 16.6        | 16.8        | 17.1        | 17.1         |
| Revenues from Carbon    | -            | -            | 0.2          | 0.9        | 2.6         | 5.5         | 9.6         | 14.3        | 19.7        | 30.0        | 37.9         |
| <b>Total revenues</b>   | <b>0</b>     | <b>0.5</b>   | <b>5.2</b>   | <b>8.8</b> | <b>14.2</b> | <b>21.5</b> | <b>25.9</b> | <b>30.9</b> | <b>36.5</b> | <b>47.1</b> | <b>55.0</b>  |
| Operating Costs (OPEX)  |              |              |              |            |             |             |             |             |             |             |              |
| Plantation              | 1.4          | 2.9          | 4.6          | 6.4        | 7.8         | 8.6         | 9.1         | 9.3         | 6.8         | 4.3         | 3.2          |
| Industry                | 0            | 0.1          | 0.7          | 0.7        | 1.3         | 1.4         | 1.5         | 1.6         | 1.6         | 1.6         | 1.7          |
| Plantation Salaries     | 0.2          | 0.2          | 0.3          | 0.4        | 0.4         | 0.5         | 0.5         | 0.6         | 0.6         | 0.6         | 0.6          |
| industry Salaries       | 0.2          | 0.3          | 0.5          | 0.6        | 0.9         | 1.0         | 1.2         | 1.3         | 1.5         | 1.6         | 1.7          |
| Depreciation            | 0.1          | 0.2          | 0.4          | 0.6        | 0.8         | 1.0         | 1.0         | 1.1         | 1.0         | 1.0         | 1.0          |
| <b>Operating Profit</b> | <b>(1.9)</b> | <b>(3.3)</b> | <b>(1.3)</b> | <b>0.1</b> | <b>3.0</b>  | <b>9.0</b>  | <b>12.6</b> | <b>17.0</b> | <b>25.0</b> | <b>38.0</b> | <b>46.8</b>  |
| <b>EBITDA</b>           | <b>(1.8)</b> | <b>(3.1)</b> | <b>(0.9)</b> | <b>0.7</b> | <b>3.8</b>  | <b>10.0</b> | <b>13.6</b> | <b>18.1</b> | <b>26.0</b> | <b>39.0</b> | <b>47..8</b> |

\*Serious Shea Senegal Key Operating Assumptions:

- Nurseries produce 1 M seedlings p.a.
- Cumulative tree planted 35M @ 80% survival rate
- Carbon revenues @ € 10 tCO2e
- 3774 T p.a afro-food processed
- CFA:USD 0.0017:1; USD:€ 1.1:1

## 7. FINANCIAL HIGHLIGHTS

### Investment Profile

| € M                                                                        | 2023         | 2024         | 2025         | 2026         | 2027       | 2028        | 2029        | 2030        | 2031        | 2032        | 2033        |
|----------------------------------------------------------------------------|--------------|--------------|--------------|--------------|------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>EBITDA</b>                                                              | <b>(1.8)</b> | <b>(3.1)</b> | <b>(0.9)</b> | <b>0.7</b>   | <b>3.8</b> | <b>10.0</b> | <b>13.6</b> | <b>18.1</b> | <b>26.0</b> | <b>39.0</b> | <b>47.8</b> |
| Working Capital                                                            | 0.1          | 0.1          | 0.4          | 0.4          | 0.4        | 0.7         | 0.4         | 0.4         | 0.5         | 0.9         | 0.8         |
| <b>Operating Cash Flow</b>                                                 | <b>(1.9)</b> | <b>(3.2)</b> | <b>(1.3)</b> | <b>0.3</b>   | <b>3.4</b> | <b>9.3</b>  | <b>13.2</b> | <b>17.7</b> | <b>25.5</b> | <b>38.1</b> | <b>47.0</b> |
| CAPEX                                                                      | 1.6          | 4.3          | 2.9          | 5.7          | 0.9        | 0.8         | 0.9         | 1.0         | 1.1         | 0           | 0           |
| Pref. Dividend                                                             | 0            | 0            | 0            | 0            | 0          | 0           | 0           | 0           | 0           | 0           | 0.000001    |
| Interest*                                                                  |              | 0.4          | 1.2          | 1.2          | 1.1        | 1.0         | 0.9         | 0.7         | 0.6         | 0.5         | 0.4         |
| Loan Repayment                                                             | 0            | 0            | 0            | 0.5          | 0.5        | 1.5         | 1.5         | 1.5         | 1.5         | 1.5         | 1.5         |
| <b>Funding requirement before Carbon Credit Prepayment Refund and Tax*</b> |              |              |              |              |            |             |             |             |             |             |             |
|                                                                            | <b>(3.5)</b> | <b>(7.9)</b> | <b>(5.0)</b> | <b>(7.1)</b> | <b>0.9</b> | <b>6.0</b>  | <b>9.9</b>  | <b>14.5</b> | <b>22.3</b> | <b>36.1</b> | <b>45.1</b> |

### Equity/Debt Profile

|                   |              |            |              |  |  |  |  |  |  |  |  |
|-------------------|--------------|------------|--------------|--|--|--|--|--|--|--|--|
| <b>MIROVA</b>     | <b>(3.5)</b> | <b>(3)</b> | <b>(3.5)</b> |  |  |  |  |  |  |  |  |
| <b>DFC /BADEA</b> |              | <b>(5)</b> | <b>(10)</b>  |  |  |  |  |  |  |  |  |

|               |            |
|---------------|------------|
| <b>Equity</b> | <b>40%</b> |
| <b>Debt</b>   | <b>60%</b> |

\* Serious Shea Senegal assumes @ 8% p.a. interest on capital debt @ 30% Tax

## 7. RISK ASSESSMENT

### Risks of high importance and mitigation approach

| Risk                                         | Description                                                                                                                                                                                                                                                                                     | Mitigation Approach                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| POLITICAL<br><b>MEDIUM</b> :                 | Presidential election changes export regulations, the investment climate and/or government commitment to the Great Green Wall Initiative and reforestation.                                                                                                                                     | Maintain a high level of integration with the national and local governments, engaging in policy and communications of the Serious Shea Project in Senegal and through processes such as the Great Green Wall Senegal Agency, African Union, UNFCCC, CBD and World Economic Forum.                                                                                                                                                                                                                                                                                                                                                     |
| CLIMATE<br><b>HIGH</b>                       | Variation in climate causing droughts, fires and flooding and variation in seasonal yields.                                                                                                                                                                                                     | Phased financing and regeneration by selection of sites with greatest potential (least degraded) and close to nurseries and communities for young tree protection and maintenance then expand the site selection outwards. Use innovative water saving technologies and recycle all water and processing and bio-gas waste for fertilizers and soil conditioners. 10% no-harvest reference areas in plantations to monitor variation against a nature and soil quality baseline and to assess nature based climate change resilience. Bush fire planning according to Senegal Code of Forestry for collaborative detection and action. |
| OPERATIONAL<br><b>MEDIUM</b>                 | Construction delays and inefficient operations which don't meet OPEX projections. Free-ranging domestic animals which damage new tree plantations.                                                                                                                                              | Significant working capital facilitates that staff can be recruited and trained and materials procured at the earliest possible date. Spiky trees are planted around the plantations as natural fences to keep out domestic animals, but also with wild animals corridors planned in collaboration with GGW Senegal.                                                                                                                                                                                                                                                                                                                   |
| SOCIO-CULTURAL<br><b>LOW</b>                 | Cooperatives refuse to collaborate with tree planting and protection and/or selling fruits, nuts and agro-forestry produce to Serious Shea.                                                                                                                                                     | The Co-operative leaders are involved in the project governance as 20% shareholders via the Community Fund managed by Golden Organics. Individuals receive a higher price than from any other buyer and can use the processing facilities for a competitive fee. Employ a specialist Stakeholder Communications Officer including to optimise input of verified local stories on social media.                                                                                                                                                                                                                                         |
| DISEASE<br><b>LOW</b>                        | Nurseries, plantations and/or farmers are plagued by pests. Supply chains are disrupted by global pandemics.                                                                                                                                                                                    | Ongoing maintenance of nurseries and plantations until trees are at least five years old and treatment with organic pesticides such as neem oil if necessary. Secured storage facilities and selection of processed product which can be stored for a year without losing quality.                                                                                                                                                                                                                                                                                                                                                     |
| FINANCE AND ESG DATA INTEGRITY<br><b>LOW</b> | A key aspect of the project is to collect and verify financial flows, production efficiency, carbon, water, energy source and efficiency and group and individual identity through the value chain. The carbon equity also relies on a market of at least €10 tCOe and accuracy of measurement. | A considerable amount of research and scoping work has created a 'beyond state of the art' Decision Support Tool to collect and verify value chain data which is then timestamped on blockchain based technologies (ChainZy) so it cannot be altered. Information, as appropriate is also calibrated with GIS and drone sources including as Level 1. collaboration with planet.com/nicfi in partnership with Google Earth Engine and UN FAO Open Foris. Deep baseline data research included carbon assessments by specialists EP Carbon.                                                                                             |



## 8. CONTACT

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**William Kwende**

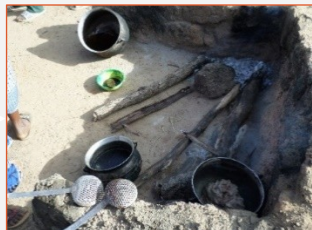
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## The Current Situation

### 1. Collection & Boiling

*Shea nuts and fuelwood are harvested and carried for miles, driving widespread deforestation, before inefficient 3-stone fires are used to boil the nuts*



### 2. Drying & Crushing

*The kernels are dried and hand-crushed, requiring long days of heavy physical labour and significant negative health impacts (e.g., smoke inhalation)*



### 3. Cooking & Extraction

*The women's groups use their hands and reused metal and plastic containers to complete the final extraction process, with limited hygiene measures*

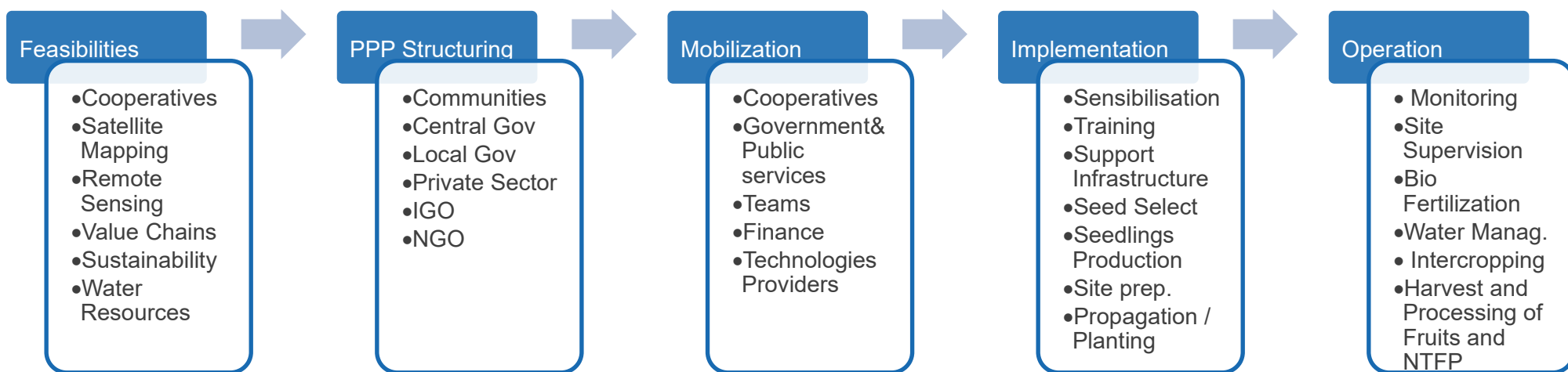


- Current agro-food processing and household cooking practices with firewood reduce the shea belt and native trees to charcoal and ashes instead of being sustainable and regenerative. For example 20kg of firewood are used to process 1kg of shea butter, driving significant deforestation and greenhouse gas emissions and requires women and children to conduct heavy physical labour and inhale smoke. Furthermore, the lack of access-to-market means that the women's groups do not receive appropriate remuneration for their input thus failing to improve their socio-economic status despite their hard labour.

**Land tenure in communal and rural domains is through cooperative instruments. Serious Shea Senegal communities have land tenure rights. Land reforestation allocation in though the Great Green Wall Senegal Agency and PAP-SEN ([papsen.org](http://papsen.org)).**



## Reforestation and Agroforestry Model



**Serious Shea Senegal's innovative approach is:**

**Science based;**

**Socially inclusive;**

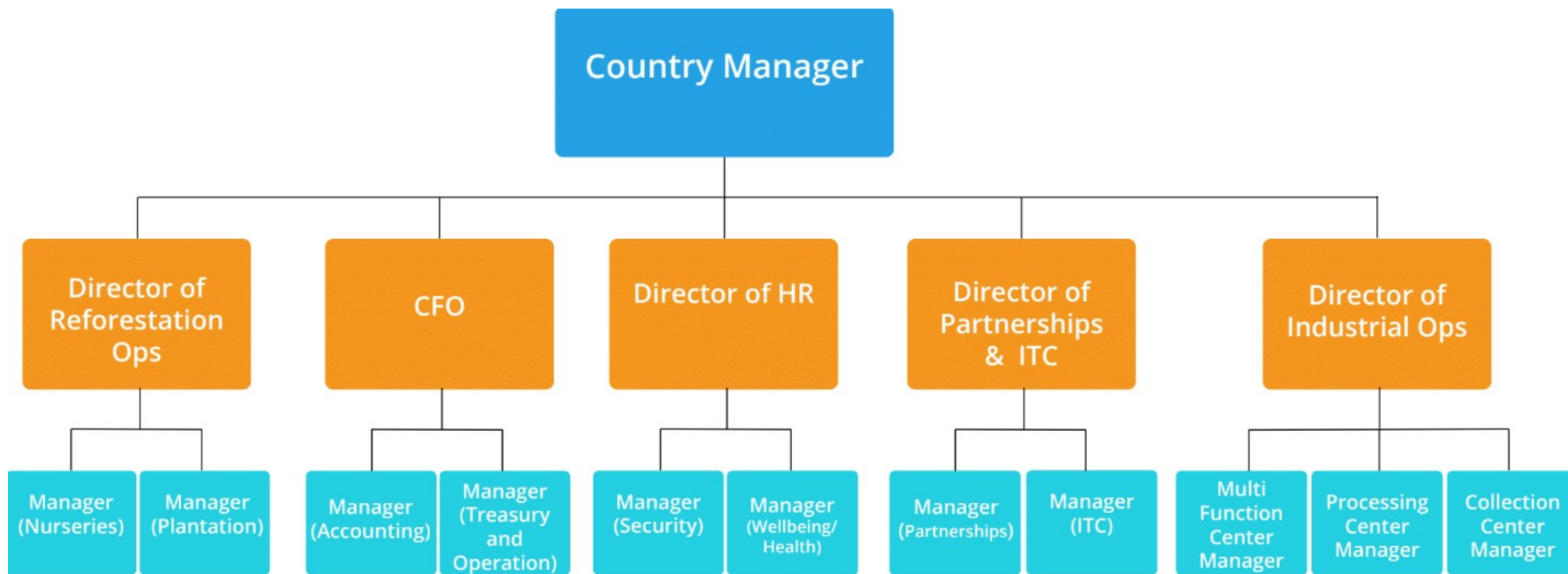
**Partnership based;**

**Built on local skills development;**

**Well managed with people and technology.**

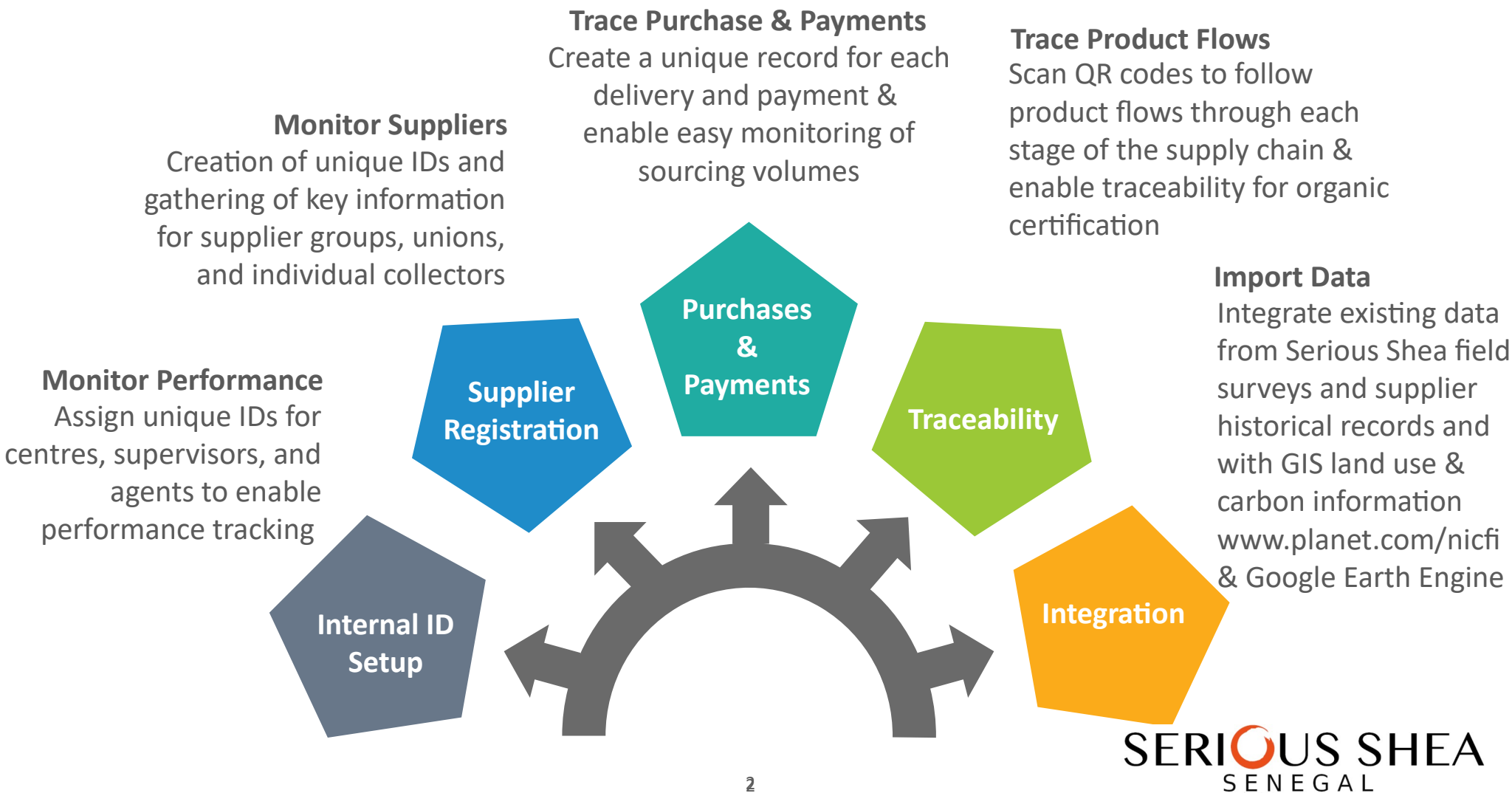


## Key Organisational Diagram

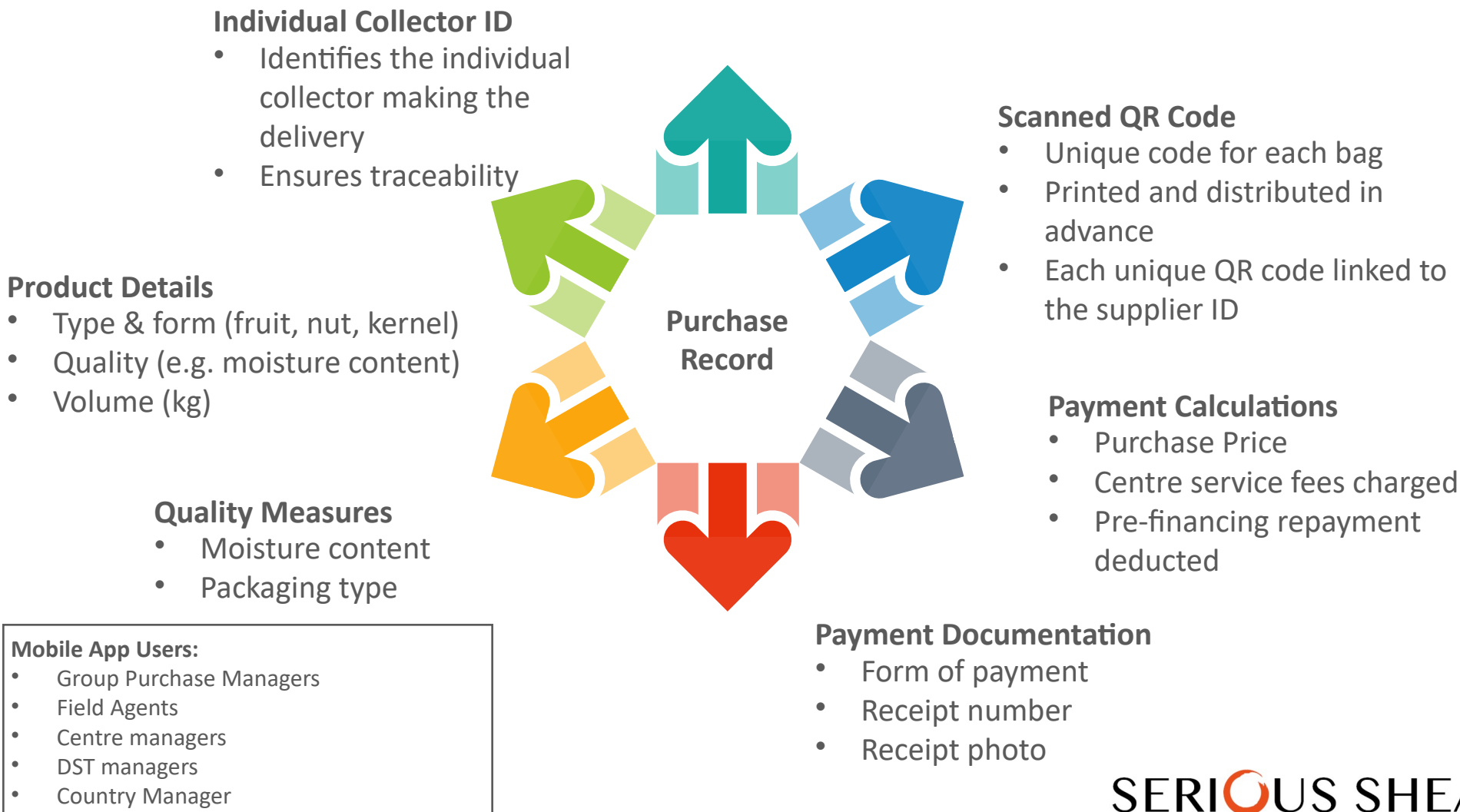


Heirarchy to 750 Staff in total for 6 plantation zones including Trainees/Students & Field Agents

## Decision Support Tool (DST) based on latest technologies

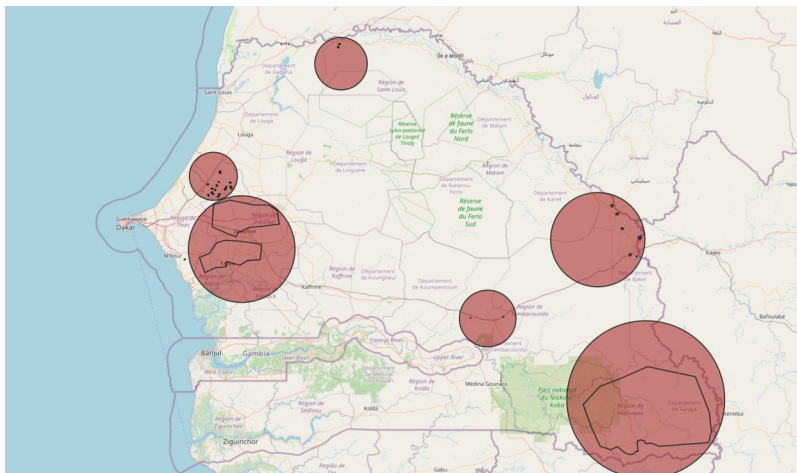


## Purchase Record - Ensuring Accountability, Transparency and Fair Pay

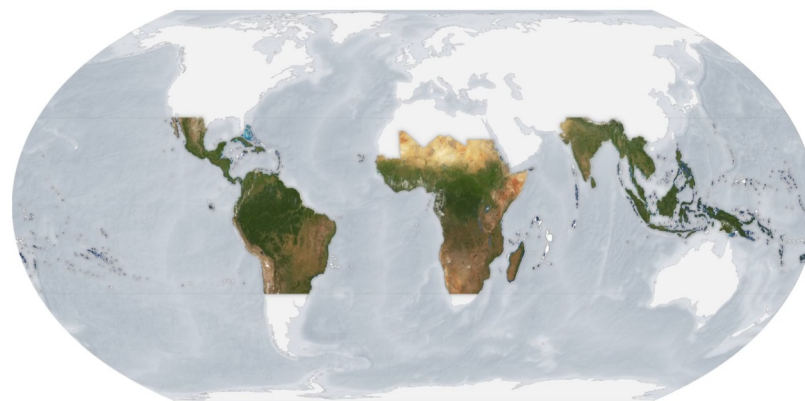


## GIS & Farm Data Interface – mapping & coraborating the latest satellite data information ‘on the farm’ & visa versa.

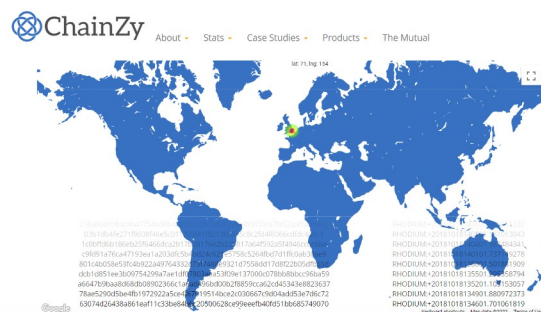
Serious Shea Senegal QGIS software for mapping individual sites with boundry geopoints for 6 Plantation Zones



Land-use and Carbon data as Seroous Shea Senegal Level 1. collaborator and user of satellite information planet.com/nicfi & google earth engine



Serious Shea Senegal farm os ULR imports data which is timestamped on Chain Zy (aka blockchain) so it cannot be altered



THE INTERNET OF RECORD  
A platform for recording internet transactions.

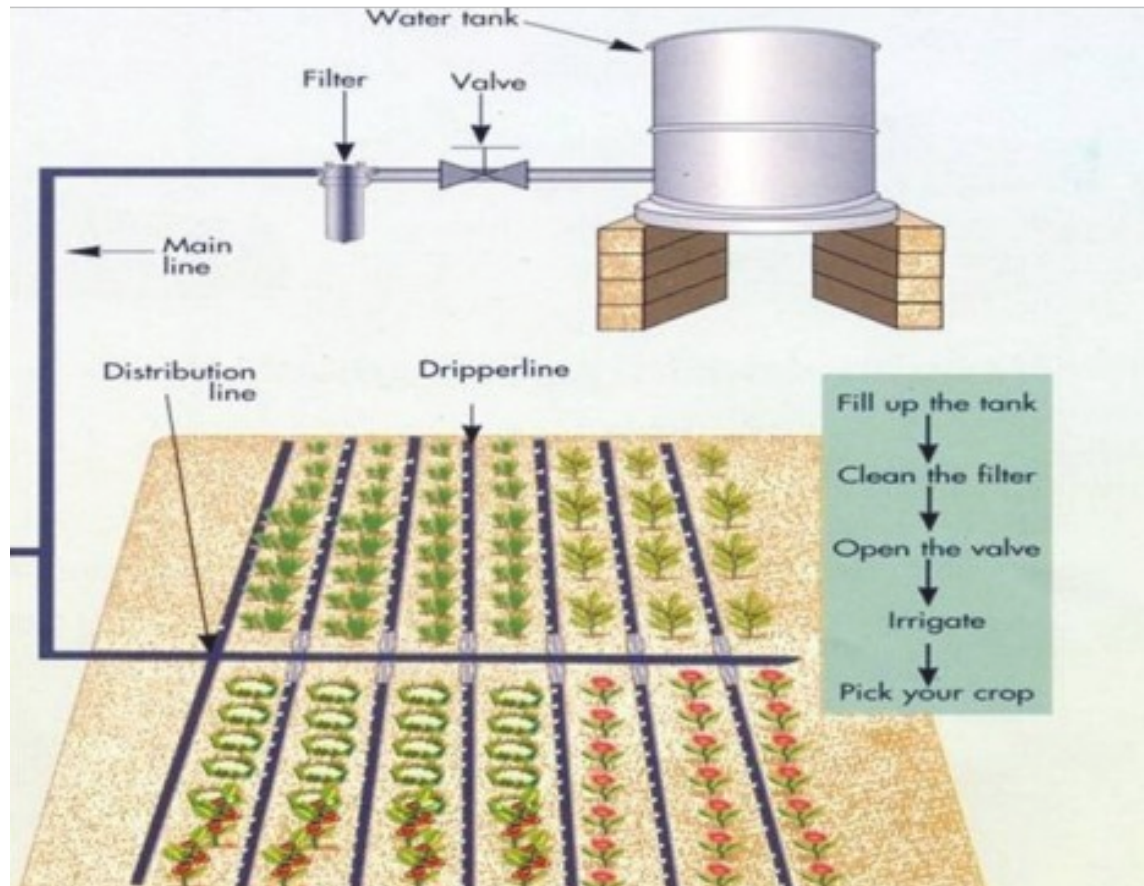


Baseline carbon data is provided in detail by Mirova Sustainable Development Fund consultants EP Carbon.

Satellite and farm data is corroborated by Serious Shea Senegal annual drone data across the 6 plantation zones.



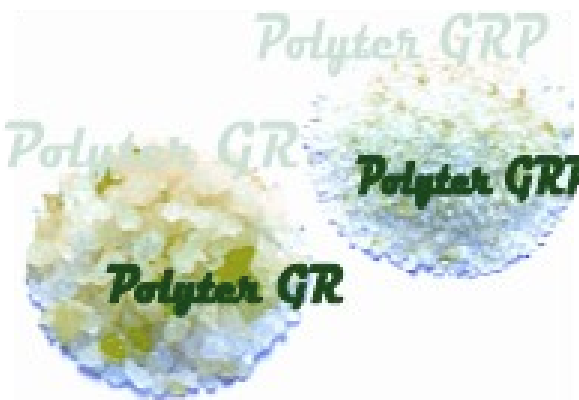
### Rural Nursery Drip Kit and Solar Pump - Adapting technology to the local environment



## Innovative Water Management and Nutrition

Gel technologies ensuring 80% survival of trees planted in Sahel conditions

- ✦ Keeps the Water and Nutrient on the root !!!
- ✦ Plants have a permanent reserve of water and nutrient
- ✦ Growth path is constant, even in desert Area ( sand)



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